Module 07: Quantization Basics

Making Models Smaller Without (Much) Pain

ECBS5200 — Applied Deep Learning
Pre-Work

What Are Model Weights?

- A trained neural network is, at the end of the day, **millions of numbers**
- These numbers are called **weights** (and biases)
- They encode everything the model learned during training
- ModernBERT-base: **~149 million** of these numbers
- Each number is stored in a specific **numerical format**

The format we choose determines **how much memory** the model needs.

Number Formats: Precision Levels

Format	Bits per Number	Bytes per Number	Nickname
FP32	32	4	Full precision
FP16	16	2	Half precision
INT8	8	1	Quarter precision

- **FP32**: The default. Maximum precision, maximum size.
- **FP16**: Half the bits. Still floating point. Very little quality loss for most tasks.
- **INT8**: Integer only. Significant size savings. Requires careful handling.

The Size Math

Take ModernBERT-base: **149M parameters**

center

Same model. Same architecture. Just stored differently.

The Trade-Off: JPEG for Models

Think of it like **JPEG compression** for images:

- Most of the time, you **cannot tell the difference** between a high-quality JPEG and the original
- But compress too aggressively, and you get **artifacts**
- The loss is usually acceptable — but **it depends on the task**

Similarly with quantization:

- FP32 → FP16: Almost always safe. Negligible quality loss.
- FP32 → INT8: Usually fine, but **measure your metrics**. Some tasks are more sensitive.
- The reconstruction error is small but **not zero**.

When Quantization Matters

Deployment & serving costs:

- A model that is 4x smaller is **4x cheaper to serve** (less GPU memory)
- Fits on **smaller, cheaper hardware**
- Lower latency — less data to move through memory

Edge & mobile:

- INT8 models can run on **phones and embedded devices**
- No cloud round-trip needed

Scale:

- Serving 1 model to 1 million users — every byte matters
- The difference between needing an A100 and getting by with a T4

What Is Coming in Week 5

This module is **intuition only**. In Week 5, you will:

- Apply **real quantization** to a trained model using industry-standard tools
- Measure the **actual impact** on your evaluation metrics
- Compare inference speed and memory usage across precision levels
- Learn about advanced techniques: dynamic quantization, quantization-aware training

For now: understand the **concept** and the **trade-off**.

Key Takeaways

1. **Model weights are just numbers** — millions of floating-point values
2. **Number format determines model size** — FP32 (4 bytes) vs INT8 (1 byte)
3. **Quantization = using fewer bits** — same model, smaller footprint
4. **4x size reduction** is typical going from FP32 to INT8
5. **Trade-off is precision vs efficiency** — like JPEG compression
6. **Measure before you ship** — always validate that quantization does not degrade your metrics below acceptable thresholds

> "The best model is the one you can actually deploy."